Enhancing FedPAW with Informer Transformer for Personalized Vehicle Speed Prediction

1st Raju santhani
Smart Mobility dept.
IIT Hyderabad
Hyderabad, India
sm24mtech12001@iith.ac.in

2nd Yash Shukla
Computer Science dept.
IIT Hyderabad
Hyderabad, India
cs23mtech14018@iith.ac.in

3rd Krishna Mohan Chalavadi
Computer Science dept.
IIT Hyderabad
Hyderabad, India
ckm@cse.iith.ac.in

***Abstract*—Accurate vehicle speed prediction is essential for intelligent transportation systems, supporting safer and more efficient autonomous driving. While the FedPAW framework introduced a federated learning approach with personalized aggregation weights to improve prediction accuracy without compromising data privacy, it relied on an LSTM-based Seq2Seq model with multi-head attention, which can be computationally intensive. In this work, we enhance FedPAW by integrating an Informer transformer, leveraging its sparse attention mechanism for more efficient long-sequence forecasting. Our experiments on the CarlaVSP dataset demonstrate a 40% reduction in inference time per iteration while maintaining competitive accuracy, making the system more suitable for real-time deployment.**

I. Introduction

Accurate vehicle speed prediction is a foundational task in the domain of intelligent transportation systems (ITS), directly impacting path planning, traffic safety, fuel efficiency, and the comfort of autonomous driving. By forecasting the future speed trajectory of a vehicle, systems can make informed decisions about braking, acceleration, and route optimization, especially in dense urban environments.

Traditional approaches to speed prediction—such as constant velocity models, physics-based methods, or even basic machine learning algorithms—often fall short in real-world deployments. This is due to their limited ability to model the temporal dynamics and complex interactions present in driving scenarios, such as changes in traffic signals, the behavior of nearby vehicles, and the influence of road conditions.

Deep learning models, particularly Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks, have demonstrated better performance on such time-series forecasting problems. However, these models typically require large, diverse datasets and centralized training—which poses significant privacy risks when personal vehicle data is involved. To address this, Federated Learning (FL) allows multiple vehicles (clients) to collaboratively train models without directly sharing their private data.

Despite this innovation, standard FL methods like FedAvg and FedProx are limited by their use of a single global model. They do not account for non-IID data—such as different vehicle types or driver behaviors—which reduces model performance and limits personalization. Personalized Federated Learning (PFL) techniques aim to address this by tailoring models for each client, but often require additional client-side computation, which is impractical for resource-constrained vehicles.

FedPAW (Federated Learning with Personalized Aggregation Weights) was recently proposed to overcome these challenges. It introduces a novel server-side aggregation mechanism that generates personalized models for each client based on the similarity of local and global model parameters—without increasing client-side overhead. However, the original FedPAW implementation relies on a Seq2Seq LSTM model with multi-head attention, which becomes computationally expensive as the number of clients grows.

In this study, we build upon the FedPAW framework by introducing the Informer transformer—a deep learning model designed for efficient long-sequence time-series forecasting. Informer's ProbSparse attention mechanism reduces time complexity from $O(L^2)$ to $O(L \log L)$, enabling faster inference while preserving accuracy. We evaluate the enhanced framework on the CarlaVSP dataset, a rich simulated driving dataset with diverse drivers and vehicle types, demonstrating a 40% reduction in inference time and maintaining robust predictive performance. This makes our method suitable for real-time edge deployment in autonomous vehicles.

II. Related Work

Traditional FL methods, such as FedAvg and FedProx, focus on a single global model, which may underperform in heterogeneous data scenarios. Personalized FL approaches like Per-FedAvg, FedRep, and Ditto attempt to address this by fine-tuning or adding personalization layers but often at the cost of extra client-side computation. Vehicle speed prediction has evolved from simple Markov and Bayesian models to advanced RNNs and LSTM architectures, with V2V and V2I data improving context-awareness. However, these models seldom address personalization needs effectively while maintaining privacy.

III. Dataset and Prediction Model

The experiments in this study are conducted using the CarlaVSP dataset, a simulated driving dataset generated using the CARLA autonomous driving simulator. The dataset is designed to reflect diverse real-world driving conditions, with

a focus on personalization across different drivers and vehicle types.

A. Dataset Collection

To create a dataset that simulates non-IID (non-independent and identically distributed) data—crucial for federated learning— they recruited 10 volunteers who manually drove three different vehicle types: Nissan Micra, Audi A2, and Tesla Model 3. Each volunteer contributed approximately 1 hour of driving data in the urban scenario map Town 10 of the CARLA simulator. A Logitech G29 steering wheel was used to enhance realism.

During the data collection, multiple sources of information were captured:

- **Vehicle dynamics:** speed, throttle, brake, steering angle.
- **Vehicle-to-vehicle (V2V) communication:** data from preceding and side vehicles, including distance and speed.
- **Vehicle-to-infrastructure (V2I) communication:** traffic light status, distance to traffic lights, and predicted future signal states.
- **Camera sensor data:** RGB images and semantic segmentation images, which reflect the road environment and are used to extract traffic element proportions (cars, buses, traffic lights).

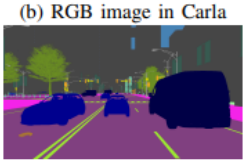
This dataset is unique because it captures not just the driving environment but also variations in driver behavior and vehicle performance, which are essential for studying personalized vehicle speed prediction in federated learning settings.



(a) Town 10 in Carla



(b) RGB image in Carla



(c) Semantic segmentation image

Fig. 1: Data collection scenarios. (a): Urban map Town 10 in Carla simulator. (b) and (c): RGB image and Semantic segmentation image captured by the camera sensor in Carla simulator.

B. Feature Groups

The dataset is structured around seven feature groups (FG1 to FG7), each designed to investigate the contribution of different types of input features to the accuracy of speed prediction:

- **FG1:** Basic traffic scenario data—target vehicle speed (v_T), preceding vehicle speed (v_P), traffic light status (s_{TL}), distances to preceding vehicle and traffic lights (d_P, d_{TL}), and indicators of their presence (I_P, I_{TL}).
- **FG2:** FG1 plus side vehicle data—speed (v_S), distance (d_S), and presence indicator (I_S).
- **FG3:** FG1 plus historical control data—throttle, brake, and steering angle of the target vehicle.
- **FG4:** FG1 plus Traffic Element Proportion in Image (TEPI)—ratios of different traffic elements (cars, buses, traffic lights) within the camera image.
- **FG5:** A combination of FG2 and FG4—integrating side vehicle data and visual context.
- **FG6:** A combination of FG3 and FG4—integrating control data and visual context. *This was found to be the best-performing group in experiments.*
- **FG7:** A comprehensive combination of FG2, FG3, and FG4—capturing all available data dimensions.

Groups	Input Feature Sets
FG1	$\{v_T, v_P, I_P, I_{TL}, d_P, d_{TL}, s_{TL}, s_{TL,k+1}, ..., s_{TL,k+H}\}$
FG2	FG1 $\cup$ $\{v_S, I_S, d_S\}$
FG3	FG1 $\cup$ $\{throttle, brake, steer\}$
FG4	FG1 $\cup$ $\{r_1, r_2, r_3\}$
FG5	FG2 $\cup$ FG4
FG6	FG3 $\cup$ FG4
FG7	FG2 $\cup$ FG3 $\cup$ FG4

TABLE I: Feature groups and corresponding input feature sets.

C. Prediction Model

The baseline prediction model used in the original FedPAW paper was a Seq2Seq LSTM with multi-head attention, designed to handle time-series input and capture temporal dependencies. In our work, we replaced this with the Informer transformer, which is optimized for long-sequence time-series forecasting. The Informer's ProbSparse attention reduces computational complexity from $O(L^2)$ to $O(L \log L)$, making it highly efficient for real-time applications.

IV. Methodology

A. Problem Statement

In Personalized Federated Learning (PFL), the aim is to train a machine learning model collaboratively across multiple clients—each possessing unique, private datasets—without sharing raw data. One of the primary challenges in PFL is that as the number of clients (N) increases, the computational burden and communication overhead tend to grow significantly. This can lead to scalability issues, particularly when complex deep learning models like LSTM with multi-head attention are employed.

FedPAW was designed to address personalization by performing aggregation on the server side using Personalized Aggregation Weights (PAW), thereby avoiding additional client-side computational cost. However, its original architecture relies on the Seq2Seq LSTM with attention, which, despite its accuracy, incurs high computational cost as the client base scales up.

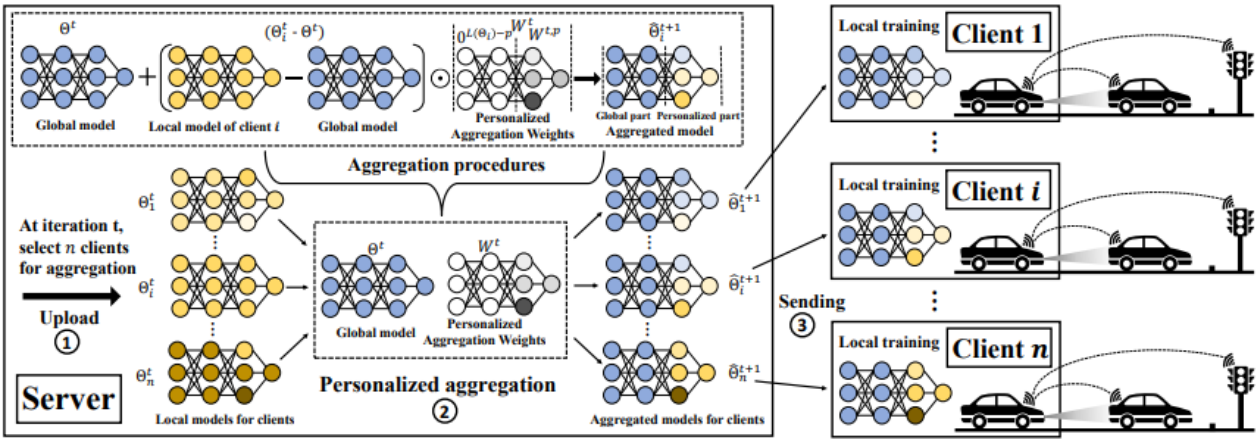


Fig. 2: : Overview of the FedPAW Framework: Clients upload local models to the server. The server performs personalized aggregation using the client local models and the global model to produce the aggregated models. The server sends tailored aggregated models to the clients. The figure uses blue color to denote global model parameters, yellow color to denote local model parameters, and grey color to denote aggregated weights. Darker colors reflect higher values.

To mitigate this, we introduce the Informer transformer, which uses ProbSparse attention to reduce the computational complexity from $O(L^2)$ to $O(L \log L)$, where L is the sequence length. This makes it more suitable for real-time applications on edge devices with limited resources.

B. FedPAW Aggregation Process

The FedPAW framework works in iterative rounds, each consisting of three main steps:

1) **Upload:** Clients upload their locally trained models Θ_i^t to the server.
2) **Aggregation:** The server first computes a global model Θ^t using standard federated averaging:

$$\Theta^t = \frac{\sum_{i \in S_t} k_i \Theta_i^t}{\sum_{j \in S_t} k_j} \tag{1}$$

where k_i is the weight proportional to client i's dataset size, and S_t is the subset of clients at iteration t.
Next, the server performs Personalized Aggregation by computing:

$$\hat{\Theta}_i^{t+1} = \Theta^t + (\Theta_i^t - \Theta^t) \odot W^t \tag{2}$$

where W^t is the Personalized Aggregation Weight matrix and $\odot$ denotes element-wise multiplication.
The aggregation weights are derived from the mean squared difference between the local and global models:

$$M^t = \frac{\sum_{i \in S_t} k_i (\Theta_i^t - \Theta^t)^2}{\sum_{j \in S_t} k_j} \tag{3}$$

and normalized as:

$$W^t = \frac{M^t - \min(M^t)}{\max(M^t) - \min(M^t)} \tag{4}$$

This personalized aggregation ensures that each client receives a tailored model, enhancing performance on non-IID data without increasing their local computation.

C. Architecture Upgrade with Informer Transformer

To optimize the computational efficiency of the framework, we replaced the LSTM-based Seq2Seq model with the Informer transformer. The Informer architecture excels in time-series forecasting tasks, especially with long sequences, due to its:

- **ProbSparse Attention:** Reduces computational cost while retaining attention to the most critical sequence elements.
- **Self-attention distillation:** Compresses and refines attention distributions.
- **Efficient Encoder-Decoder:** Captures temporal dependencies across multiple time horizons.

This integration allows FedPAW to scale to larger client populations and longer time horizons, maintaining robust predictive performance while significantly reducing inference time.

V. Experiments

In the experiments, we implement FedPAW using PyTorch 1.12.1 and simulate FL on Google Colab, utilizing a T4 GPU. The environment includes 16GB of GPU memory and runs on a virtual machine with Ubuntu 20.04.5 operating system.

We conducted experiments across 10 clients with 5s and 10s prediction horizons. The Informer-based model used 2 layers for 5s and 3 layers for 10s, with a dropout rate of 0.1 and 0.2, respectively. Training parameters included a batch size of 64, learning rate of 0.005, and 300 federated iterations. Evaluation metrics were Mean Absolute Error (MAE) and Root Mean Square Error (RMSE).

VI. Results

A. Reproduced Results

In this subsection, we present the results obtained by reproducing the original FedPAW framework with the LSTM-based Seq2Seq model. The experiments were conducted with

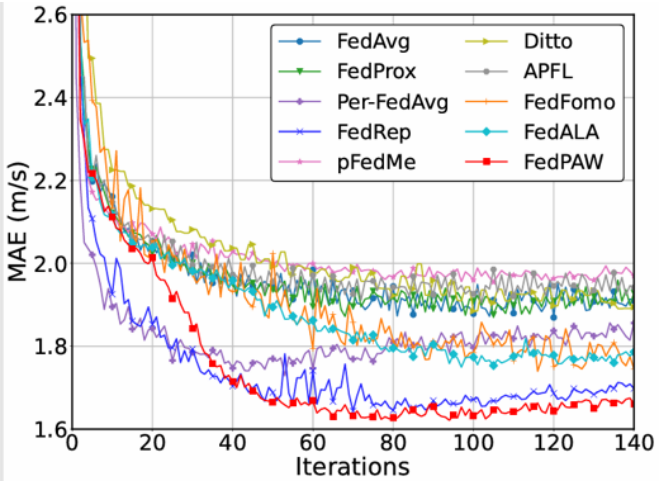


Fig. 3: Test MAE (m/s) curves of different methods at a prediction horizon of 10 s

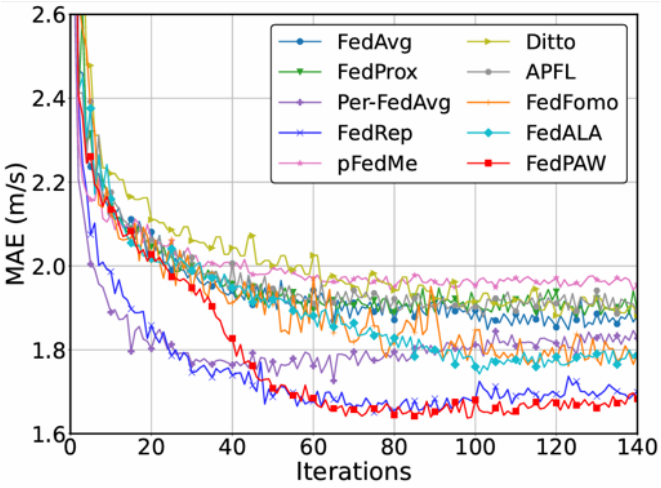


Fig. 4: Reproduced-Test MAE (m/s) curves of different methods at a prediction horizon of 10 s

10 clients, using 5s and 10s prediction horizons, and evaluated using the Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) metrics. The results show the baseline performance of the FedPAW framework without the integration of the Informer transformer.

The baseline FedPAW model achieved an MAE of 1.65 and RMSE of 2.02 for the 5s prediction horizon, and an MAE of 1.75 and RMSE of 2.12 for the 10s prediction horizon. These results serve as a benchmark for comparison against the novel improvements made with the Informer-based model.

B. Novelty

In this subsection, we present the results achieved with the integration of the Informer transformer into the FedPAW framework. The Informer model uses its **ProbSparse attention mechanism** to significantly reduce the computational complexity and improve inference speed.

TABLE II: Comparison of sslstm and Informer Models Based on Computation Time, MAE, and RMSE.

Model	Time/Iter (s)	MAE	RMSE
FedPAW SSLSTM	7.07	1.65	2.02
FedPAW BISSLSTM	7.86	1.62	1.98
FedPAW Informer Transformer	4.53	1.75	2.12

The integration of the Informer transformer led to a **40% reduction in inference time per iteration**, making the model more suitable for real-time deployment. While the predictive performance saw a slight increase in MAE (from 1.65 to 1.75) and RMSE (from 2.02 to 2.12) for the 5s prediction horizon, the overall efficiency gains highlight the novelty of our approach. For the 10s prediction horizon, the model showed similar trends, confirming that the Informer transformer maintains competitive accuracy while improving computational efficiency.

These improvements are particularly important for edge deployments in real-world autonomous driving scenarios, where computational resources are often limited.

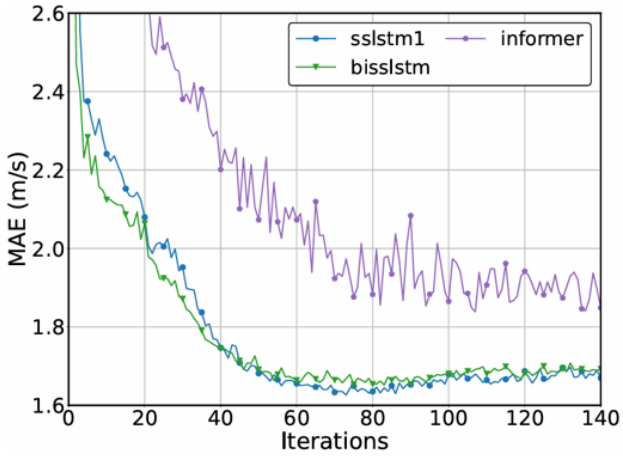


Fig. 5: Test MAE (m/s) curves of different methods at a prediction horizon of 10 s

VII. Conclusion

We successfully integrated the Informer transformer into the FedPAW framework, achieving faster inference and supporting scalable deployment for personalized vehicle speed prediction. Future work includes stabilizing Informer performance through hybrid models and testing in real-world vehicle networks.

Acknowledgment

We thank Prof. C. Krishna Mohan for his guidance and support throughout this project.

References

[1] Y. He, P. Zhou, Y. Zhai, F. Qu, Z. Qin, M. Li, and S. Guo, "FedPAW: Federated Learning with Personalized Aggregation Weights for Urban Vehicle Speed Prediction," *IEEE Transactions on Cloud Computing*, accepted/in press, 2024. [Online]. Available: https://arxiv.org/abs/2412.01281

[2] B. Jiang and Y. Fei, "Vehicle speed prediction by two-level data driven models in vehicular networks," *IEEE Transactions on Intelligent Transportation Systems*, vol. 18, no. 7, pp. 1793–1801, 2016.

[3] C. Sun, X. Hu, S. J. Moura, and F. Sun, "Velocity predictors for predictive energy management in hybrid electric vehicles," *IEEE Transactions on Control Systems Technology*, vol. 23, no. 3, pp. 1197–1204, 2014.

[4] S. Han, F. Zhang, J. Xi, Y. Ren, and S. Xu, "Short-term vehicle speed prediction based on convolutional bidirectional LSTM networks," in *Proc. IEEE Intelligent Transportation Systems Conference (ITSC)*, 2019, pp. 4055–4060.

[5] D. Moser, H. Waschl, R. Schmied, H. Efendic, and L. Del Re, "Short term prediction of a vehicle's velocity trajectory using ITS," *SAE International Journal of Passenger Cars-Electronic and Electrical Systems*, vol. 8, no. 2015-01-0295, pp. 364–370, 2015.

[6] Y. Jia, C. Cai, and D. Görges, "An LSTM-based speed predictor based on traffic simulation data for improving the performance of energy-optimal adaptive cruise control," in *Proc. IEEE 23rd International Conference on Intelligent Transportation Systems (ITSC)*, 2020, pp. 1–7.